

TRAN VAN MANH

Robotics Engineer | Motion Planning & Deep Reinforcement Learning

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[LinkedIn](#) • [Google Scholar](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

I am a Robotics Engineer with more than 5 years of experience in autonomous navigation systems, motion planning, SLAM & localization, and AI-driven robotic applications. I enjoy exploring new technologies, tackling challenging engineering problems, and transforming research ideas into real-world robotic solutions. Skilled in designing and deploying ROS/ROS2-based systems with behavior-tree decision-making, global and local path planning, and real-time motion control on embedded/edge Linux platforms, with a proven record of transferring reinforcement-learning-based (Physical AI) navigation policies from simulation to real-world robots (sim-to-real).

TECHNICAL SKILLS

Core Expertise: Autonomous Navigation, Mobile Robotics, Motion & Path Planning, Sampling-based & Search-based Planning (A*, Hybrid-A*, RRT), Deep Reinforcement Learning (Physical AI), Sim-to-Real Transfer, Multi-Robot Systems, Probabilistic Robotics

Programming: C++, C, Python, MATLAB

Frameworks & Libraries: ROS / ROS2, Nav2, PyTorch, OpenAI Gym, OpenCV, RTAB-Map, Matplotlib

Control: MPPI, MPC, PID, Backstepping, Sliding-Mode Control, Kalman Filter

Simulation & Tools: Gazebo, MuJoCo, Stage, V-REP, Git / GitHub, LaTeX

Embedded: STM32 (ARM)

PROFESSIONAL EXPERIENCE

Robotics Engineer — Robot Global Team (RGT)

Sep 2024 – Present

Local Planning & Behavior Trees for Service Robots Seoul, South Korea

- Developed and optimized local planners (TEB, MPC, and MPPI) in Nav2 for autonomous navigation of service and multi-robot systems, reducing on-board computation cost on edge Linux single-board computers (SBCs) by 30% to ensure real-time performance and stable control-loop timing.
- Implemented full autonomous navigation with Nav2 and designed behavior-tree decision-making with BehaviorTree.CPP for complex scenarios (e.g., narrow-path waiting logic).
- Supported the development of multi-floor navigation and docking features for service robots.
- Developed and maintained the motor controller (emergency-stop logic, robust serial communication, safety handling, battery-voltage filtering and logging); released production software deployed in restaurant environments.

Researcher — Intelligent Robot Laboratory, CBNU

Mar 2022 – Aug 2024

ATC Project: Deep-RL Collision-Avoidance Navigation Cheongju, South Korea

- Successfully transferred deep reinforcement learning navigation models from simulation to real-world service robots (sim-to-real).
- Evaluated mapless navigation and localization with a motion-capture system, achieving a >95% navigation success rate and <10% localization error; performance certified under Korean (KR) safety standards.

Researcher — Vietnam-Korea Institute of Science & Technology (VKIST)

Aug 2020 – Feb 2022

KIST Sub-project — Industrial AMR Hanoi, Vietnam

- Researched and developed autonomous mobile robot (AMR) technology for industrial transport applications.
- Defined functions, requirements, specifications, and technical criteria, and designed the kinematic model and autonomous navigation framework for an omnidirectional mobile-robot prototype.
- Designed and optimized magnetic flux density of a PMSM motor in JMAG; applied genetic-algorithm optimization, the ROS navigation stack, LiDAR-based localization (AMCL), and Kalman filtering.

Research Intern — Vietnam Academy of Science & Technology (VAST)

Mar 2018 – Aug 2020

Project: "Smart Human-Form IVASTBot Robot" (Phase 01) Hanoi, Vietnam

- Designed an intelligent trajectory-tracking system for an omnidirectional mobile robot.
- Developed backstepping and sliding-mode control algorithms in MATLAB and deployed low-level control in C on embedded systems.

EDUCATION

M.Sc. in Control and Robot Engineering

Mar 2022 – Aug 2024

Chungbuk National University (CBNU) Cheongju, South Korea

- GPA: 3.69 / 4.0 (4.19 / 4.5).
- **Thesis:** *Autonomous Navigation in Complex Environments Based on Deep Reinforcement Learning Using Selective Policy Switching for Mobile Robots.*

B.Eng. in Control Engineering and Automation

Aug 2015 – Aug 2020

Hanoi University of Science and Technology (HUST) Hanoi, Vietnam

- GPA: 3.12 / 4.0.
- Ranked 1st in Vietnam for Electrical and Electronic Engineering (QS World University Rankings).

PUBLICATIONS & PATENTS

Journal Articles

- **Tran, V. M.** and Kim, G.-W. (2024). "Cooperative Deep Reinforcement Learning Policies for Autonomous Navigation in Complex Environments." *IEEE Access*, vol. 12, pp. 101053–101065. **(First author)**
- **Tran, V. M.** and Kim, G.-W. (2024). "Mapless Navigation with Distributional Reinforcement Learning." *Journal of Korea Robotics Society*, vol. 19(1), pp. 92–97. **(First author — Best Paper Award)**
- Ha Thi, K. D., **Tran, V. M.**, et al. (2021). "Adaptive Control for Uncertain Model of Omnidirectional Mobile Robot Based on Radial Basis Function Neural Network." *International Journal of Control, Automation, and Systems*, vol. 19, pp. 1715–1727. (Co-author)
- Ha Thi, K. D., **Tran, V. M.**, et al. (2020). "Autonomous Navigation for Omnidirectional Robot Based on Deep Reinforcement Learning." *International Journal of Mechanical Engineering and Robotics Research*. (Co-author)

Conference Papers

- **Tran, V. M.**, et al. (2019). "Mapping and Navigation with a Four-Wheeled Omnidirectional Mobile Robot Based on the Robot Operating System." *MoRSE*, Bali, Indonesia. (Co-author)
- Ha Thi, K. D., **Tran, V. M.**, et al. (2019). "Trajectory Tracking Control for a Four-Wheeled Omnidirectional Mobile Robot Using Backstepping Aggregated with Sliding Mode Control." *ICA-SYMP*. (Co-author)
- Nguyen, T. L., **Tran, V. M.**, et al. (2019). "Nonlinear Backstepping-Sliding Mode Control of Electro-Hydraulic Systems." *ICERA*, Springer. (Co-author)
- Nguyen, M. C., **Tran, V. M.**, et al. (2019). "Neural-Network-Based Adaptive Control of Web Transport Systems." *ICSSE*, pp. 124–128. (Co-author)

Patents

- **Tran, V. M.** and Kim, G.-W. (2024). "Autonomous Navigation System in Complex Environments Based on Deep Reinforcement Learning Using Selective Policy Switching for a Mobile Robot." KR Patent MP24-0134 / MP24-0141.

HONORS & AWARDS

- Best Paper Award, Journal of Korea Robotics Society (2024).
- Korean (KR) Safety Certification for human-robot navigation performance (2024).
- Graduate Scholarship for four semesters of the M.Sc. program, Chungbuk National University (2022–2024).
- Best Graduation Thesis Defense, Hanoi University of Science and Technology (2020).

LANGUAGES

Vietnamese (Native) • English (TOEIC 740 / IELTS 6.0)